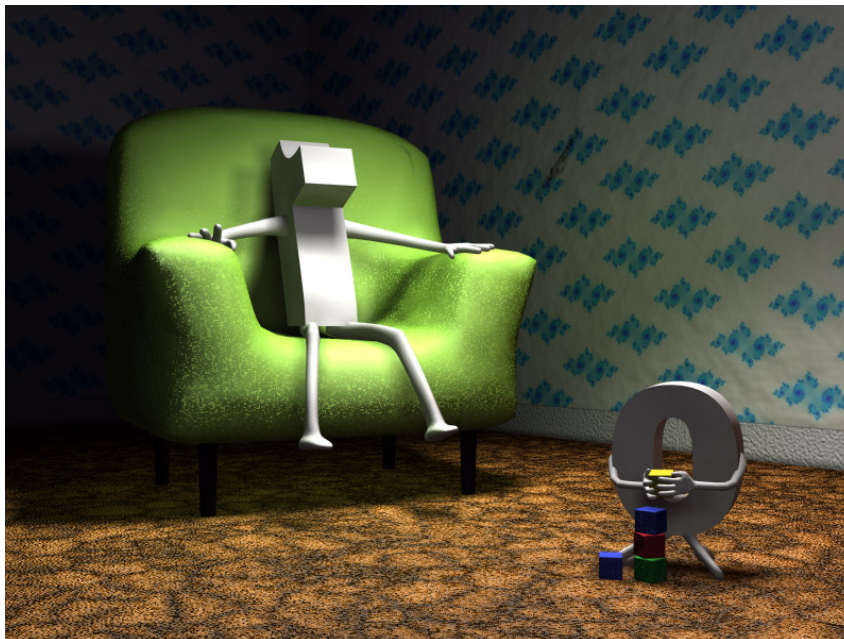


Chapter 1. INTEGERS

1. 3 Integers modulo n



Today we will introduce an equivalence relation on $\mathbb{Z}$, called **CONGRUENCE**. We will

- Prove very important properties of such relation
- Take a very closer look to the equivalent classes

Congruence modulo n is an equivalence relation } THEM 1.32

Definition 1.27. Let $n \geq 2$, a and b be integers. We say that a and b are **congruent modulo n** if $n|(a - b)$. In such a case, we write $a \equiv b \pmod{n}$ and refer to n as the **modulus**.

Example 1.29. $2 \equiv 5 \pmod{3}$, $21 \equiv 16 \pmod{5}$, $-4 \equiv 2 \pmod{6}$.

$\underbrace{2 - 5 = -3}$
 and $3|(-3)$

$\underbrace{21 - 16 = 5}$
 and $5|5$

$\underbrace{-4 - 2 = -6}$
 and $6|(-6)$

What are we really doing here?



Think about a clock that has struck 3 p.m.

The time after 10 hours will be 1 a.m. We don't say that it is 13 ...

Using the congruence modulo $n = 12$, we have that

$$13 \equiv 1 \pmod{12}$$

We will say that the residue class of 13 modulo 12 equals to the residue class of 1 modulo 12. We will talk about this now.

Definition 1.33. The equivalence class $[a]_n$ or $[a]$ of an integer a with respect to congruence modulo n is called its **residue class modulo n** . It consists of the integers whose remainder is a when divided by n :

$$[a]_n = \{x \in \mathbb{Z} \mid x \equiv a \pmod{n}\}.$$

We say that a is a generator of $[a]_n$. Notice that $[a]_n = [b]_n$ if and only if $a \equiv b \pmod{n}$. The set consisting of all the residue classes modulo n is denoted by $\mathbb{Z}_n$, and is called the **set of integers modulo n** .

THEOREM 1.34. Let $n \geq 2$ be an integer.

- (1) If $a \in \mathbb{Z}$, then $[a]_n = [r]_n$ for some $0 \leq r \leq n-1$.
- (2) The residue classes $[0]_n, [1]_n, \dots, [n-1]_n$ are all distinct.

Corollary 1.35. $\mathbb{Z}_n$ has n elements: $\mathbb{Z}_n = \{[0]_n, [1]_n, \dots, [n-1]_n\}$

Proof.

(1) For $a \in \mathbb{Z}$, apply the Division Algorithm to write

$$a = qn + r, \text{ where } q \in \mathbb{Z}, 0 \leq r \leq n-1.$$

Then $a - r = qn$, which says that $n \mid (a - r)$.

Thus: $[a]_n = [r]_n$.

We have proved that the equivalence class of a

coincides with the equivalence class of the remainder when divided a by n .

(2) Suppose that $[r]_n = [s]_n$ for some $0 \leq r, s \leq n-1$. We can assume without loss of generality that $r \leq s$.

$$[r]_n = [s]_n \Rightarrow s \equiv r \pmod{n} \Rightarrow n \mid (s-r)$$

$\Rightarrow s-r$ is a multiple of n

$$\Rightarrow s-r \geq n$$

But both $0 \leq r, s \leq n-1$, so this is impossible. $\blacksquare$

$$\mathbb{Z}_n = \left\{ [0]_n, [1]_n, \dots, [n-1]_n \right\}$$

Example 1.36. $\mathbb{Z}_2$ has two elements, which are the residue classes of the remainders obtained when divided by 2. There only two possible remainders, namely, 0 and 1. So, $\mathbb{Z}_2 = \{[0]_2, [1]_2\}$, where

$[0]_2 = \{x \in \mathbb{Z} \mid x \equiv 0 \pmod{2}\}$ = the set of even integers

$[1]_2 = \{x \in \mathbb{Z} \mid x \equiv 1 \pmod{2}\}$ = the set of odd integers

Example 1.37. Find $[48]$ and $[-16]$ in $\mathbb{Z}_7$.

$$\mathbb{Z}_7 = \{[0], [1], [2], [3], [4], [5], [6]\}$$

$$\overset{?}{[48]} \in \mathbb{Z}_7$$

$$48 = 6 \cdot 7 + \underline{6}$$

The equivalence class of $[48]$ in $\mathbb{Z}_7$ is the equivalence class of the remainder when divided 48 by 7.

$$[48] = [6] \text{ in } \mathbb{Z}_7.$$

$$[-16] = ? \text{ in } \mathbb{Z}_7$$

$$-16 = (-3)7 + \underline{5}$$

remainder

$$[-16] = [5] \text{ in } \mathbb{Z}_7$$

